

Individual Project Report

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[GitHub Repo](#)

Abstract

This Project asks a straightforward question “what is the major factor that affects the energy cost of a household in the USA ?” and uses statistics , both descriptive and inferential to find a reasonable answer. In order to perform this statistical analysis a government produced dataset is used (RECS) consisting of ~18500 households with a detailed array of 799 features. However, The features that we explored were just a small subset, namely :

Feature	Description
DOEID	Unique household ID
TOTALDOL	Annual energy cost (all fuels)
TOTALBTU	Annual site energy
TOTSQFT_EN	Energy-used floor area
HDD65	Heating degree days
CDD65	Cooling degree days
REGIONC	Census region
TYPEHUQ	Housing type
YEARMADERANGE	Year built (banded)

Moreover, all the hypotheses that were used during this research were focussed on these features and how they affected the final Annual energy cost. Only a small amount of features were selected because of their quality and relevance to the Business Problem. a bigger number of dataset would have resulted in a more in-depth analysis

1 Business Problem

The main question that is posed here is : **What is the main reason of high energy consumption for a household in the USA ?**

This problem is relevant to multiple parties. A utility, planning to create new plants or programmes for existing customers to increase efficiency . Policymakers trying to allocate financial and human resources at the cause that is affecting energy cost the most as well as the homeowners themselves trying to budget their fixed costs or looking for a new house and considering energy cost as a big factor.

The objective of this analysis would be to find a statistically right answer for this question, backed by data, something that can produce real outcomes if acted upon.

2 Exploratory Data Analysis

The dataset has in total 18495 rows which are individual households recorded in the survey that created this dataset. performing a View() on the dataframe shows us that there are 799 features in this dataset.

To understand the contents of these features as well as detecting null values , a new view called inventory was created

	variable	type	n_distinct	n_missing
1	DOEID	integer	18496	0
2	REGIONC	character	4	0
3	DIVISION	character	10	0
4	STATE_FIPS	integer	51	0
5	state_postal	character	51	0
6	state_name	character	51	0
7	BA_climate	character	8	0
8	IECC_climate_code	character	17	0
9	UATYP10	character	3	0
10	HDD65	integer	6880	0
11	CDD65	integer	3909	0

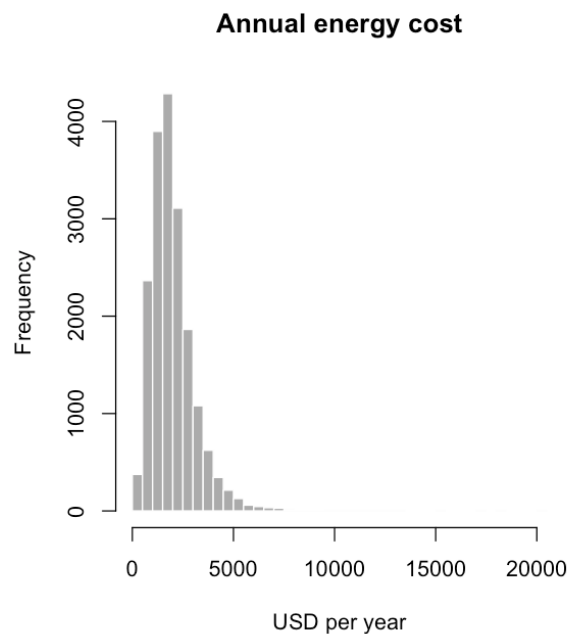
This view shows us the number of distinct values per feature and if they have any explicit null values.

Next, after going through the documentation of the dataset which explains the meaning of the names of the features(as they are very descriptive here in the dataset). Only 9 of them were selected for the analysis because they were the most relevant features to our business problem.

Our Target variable being TOTALDOL which is the annual energy cost for all fuels

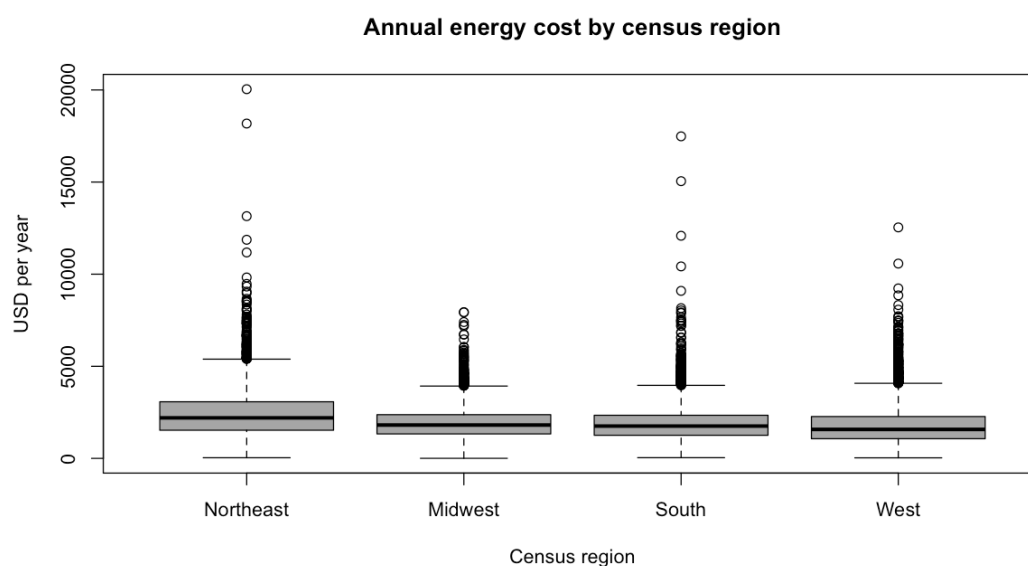
combined. here is the code for a visualization of this feature describing how many of the households have what cost in USD per year.

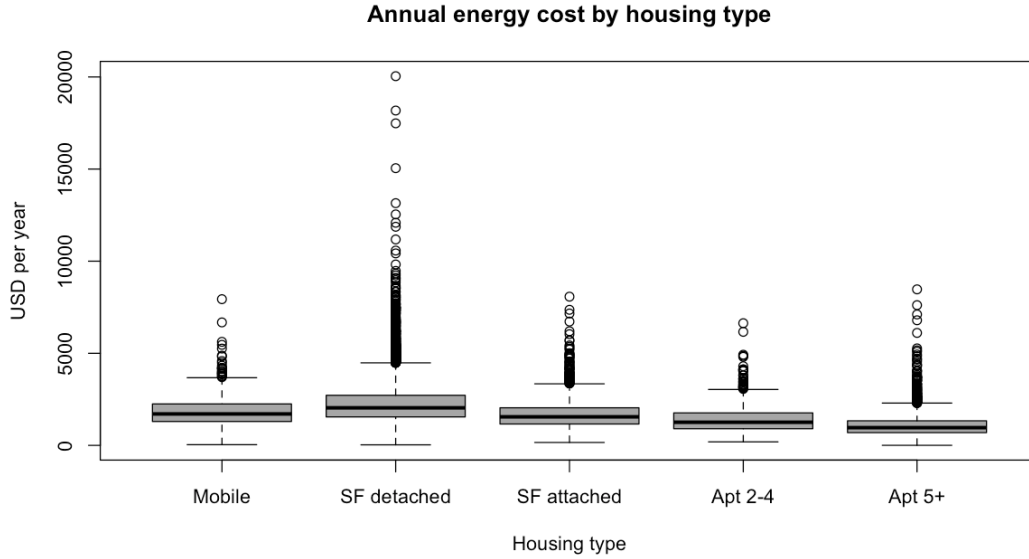
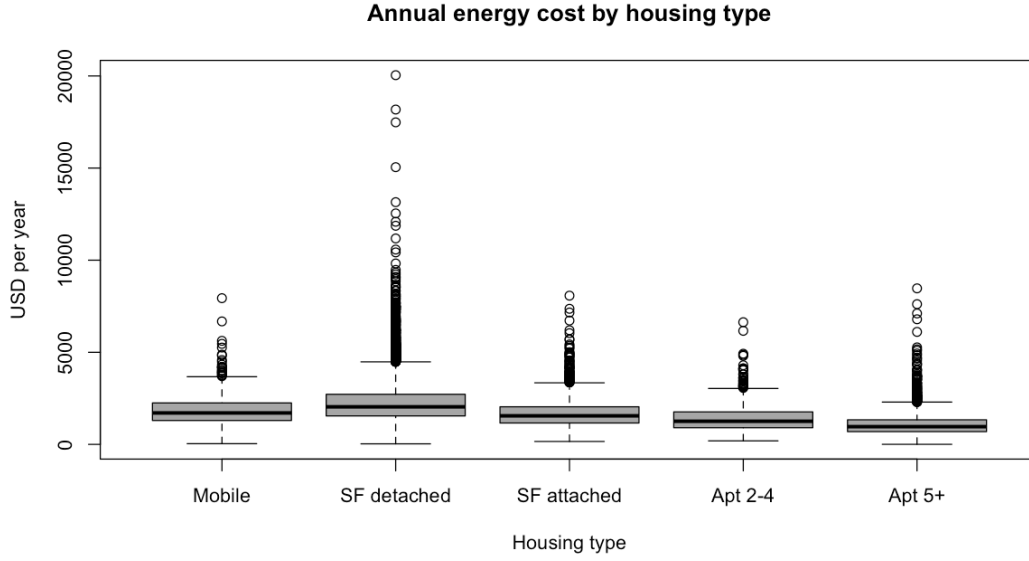
```
old_par <- par(mfrow = c(1, 2))
hist(recs$TOTALDOL, breaks = 60, col = "grey67", border = "white",
      main = "Annual energy cost", xlab = "USD per year")
```



As you can see the energy cost is heavily right skewed meaning most households have relatively less energy cost per year whereas some households have extremely high.

here are some other visualizations that are interesting first findings, all are boxplots.





3 Hypotheses to be Tested

The following. are the 4 Hypotheses that are to be tested in this research project :

- Homes built before 2000 and from 2000 onward have equal mean annual cost
- Mean annual cost is equal in all four census regions
- Floor area has no effect on cost
- Floor area and the climate measures have equal or no effect.

4 Data Preprocessing

The first decision as a pre-processing step was to only choose 9 of the 799 features:

```

56 # Choosing the variables needed for the analysis.
57 vars_needed <- c(
58   "DOEID",      # unique household id
59   "TOTALDOL",   # annual energy cost for all fuels
60   "TOTALBTU",   # annual site energy
61   "TOTSQFT_EN", # energy-used floor area
62   "HDD65",      # heating degree days
63   "CDD65",      # cooling degree days
64   "REGIONC",    # census region
65   "TYPEHUQ",    # housing type
66   "YEARMADERANGE" # year built, banded
67 )
68
69 recs_raw <- recs_full[, vars_needed]
70
71 cat("\nStructure of the data:\n")
72 str(recs_raw)

```

Second, a cleaning step was performed that was due to a certain limitation of the dataset mentioned in the dataset documentation, which says that values that are not applicable to a certain household is recorded as -2 . This would mess up all the numeric calculations. A check was performed if such values persist in the chosen features along with detecting null values in them

```

75 # SECTION 2 - CHECK AND CLEAN
76
77 # the documentation of the dataset explains that RECS codes saves "not
  applicable" as -2, which read.csv imports as a normal number, checking if
  our respective features have this value, also checking for nulls.
78
79 checks <- data.frame(
80   n_missing = vapply(recs_raw, function(x) sum(is.na(x)), integer(1)),
81   n_code_na = vapply(recs_raw, function(x) {
82     if (is.numeric(x)) sum(x == -2, na.rm = TRUE) else 0L
83   }, integer(1))
84 )
85
86 cat("\nCompleteness check:\n")
87 print(checks)
88
89 |
90 # Cost must be positive
91
92 keep <- recs_raw$TOTALDOL > 0 & recs_raw$TOTSQFT_EN > 0
93 recs <- recs_raw[keep, ]

```

```

> print(checks)
              n_missing n_code_na
DOEID              0          0
TOTALDOL            0          0
TOTALBTU            0          0
TOTSQFT_EN          0          0
HDD65               0          0
CDD65               0          0
REGIONC            0          0
TYPEHUQ             0          0
YEARMADERANGE       0          0

```

Because no such value exists, no need to drop or amputate any rows.

Furthermore, The features REGIONC which has data about different regions was changed from character array to a factor variable type so that it becomes categorical and can be used in our analysis. The same was done for TYPEHUQ and YEARMADERANGE.

A train/test sample of an 80/20 split was created in order to test the 2 regression models

5 Inferential Statistical Tests

- For Homes built before 2000 and from 2000 onward have equal mean annual cost:
 - Welch two-sample t-test would work the best here as 2 groups are being compared here.
- Mean annual cost is equal in all four census regions
 - One-way ANOVA was selected because there are 4 different type of groups and according to the test flavour chart, this would be the most appropriate test.
- Floor area has no effect on cost
 - A Simple Linear Regression was chosen with the slope tested by a t-test on the coefficient, in order to find out the effect of Floor area to the cost, assuming a linear relationship and trying to predict the annual bill of a house based on the floor area.
- Floor area and the climate measures have equal or no effect.
 - A Multiple linear regression was used with the joint null tested by the model F-test because of 2 predictors are used simultaneously, and by this the coefficient can be adjusted for one by the other.

6 Applying Statistical Test

Applying the Welch two sample t-test

```

220 # Assumption 1: unequal variances. Bartlett's test.
221 cat("\nBartlett test:\n")
222 print(bartlett.test(TOTALDOL ~ vintage, data = recs))
223
224 # Assumption 2: normality. QQ plot per group, plus Shapiro-Wilk on a random
225 # subsample (the test accepts at most 5000 values).
226 old_par <- par(mfrow = c(1, 2), mar = c(6, 4.5, 4, 2))
227 for (g in levels(recs$vintage)) {
228   qqnorm(recs$TOTALDOL[recs$vintage == g],
229         main = paste("Homes built", g),
230         xlab = "Theoretical quantiles (normal)",
231         ylab = "Observed annual cost (USD)")
232   qqline(recs$TOTALDOL[recs$vintage == g])
233   mtext("Points on the line = normally distributed",
234         side = 1, line = 4.2, cex = 0.75)
235 }
236 par(old_par)
237

```

gives us this result :

```

220 # Assumption 1: unequal variances. Bartlett's test.
221 cat("\nBartlett test:\n")
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233   mtext("Points on the line = normally distributed",
234         side = 1, line = 4.2, cex = 0.75)
235 }
236 par(old_par)
237

```

Welch Two Sample t-test

```

**t == 3.4275, **df == 7463.5, **p-value == 0.0006125

```

```

**alternative hypothesis:**true difference in means between group Before
    2000 and group 2000 or later is not equal to 0.

```

```

**sample estimates:**

```

```

mean in group Before 2000 mean in group 2000 or later

```

```

2006.425          1940.383

```

```

**Difference in means:** 66.04 USD per year (3.3% of the mean bill)

```

```

Null Hypothesis rejected, but not a significant difference found

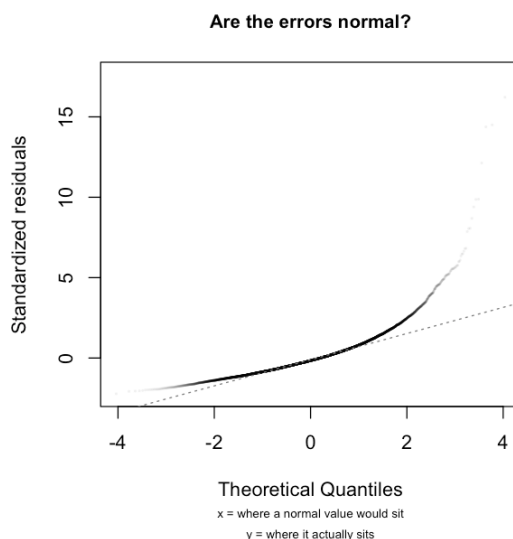
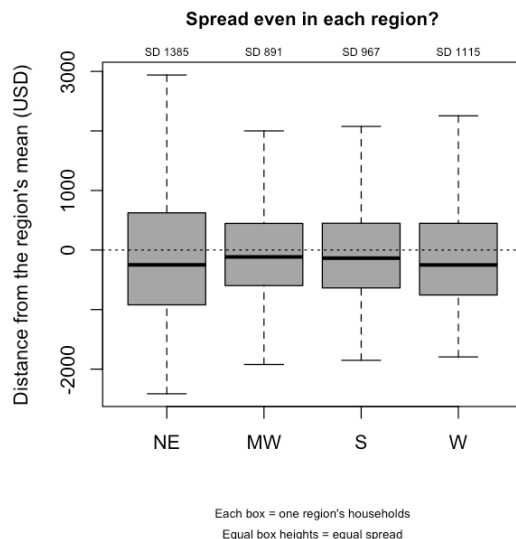
```

Applying the one-way ANOVA

```

277
280 # SECTION 5 ONE-WAY ANOVA ACROSS REGIONS
281
282 # H0: mean annual energy cost is equal in all four census regions.
283 # H1: It differs
284
285 cat("\nGroup descriptives:\n")
286 print(describeBy(recs$TOTALDOL, recs$REGIONC, mat = TRUE)[
287   , c("group1", "n", "mean", "sd", "se")])
288
289 aov_region <- aov(TOTALDOL ~ REGIONC, data = recs)
290
291 cat("\nOne-way ANOVA:\n")
292 print(summary(aov_region))
293
294 # Assumption 1: equal variances across groups.
295 # Assumption 2: normally distributed Residual values (Variances)
296 cat("\nBartlett test:\n")
297 print(bartlett.test(TOTALDOL ~ REGIONC, data = recs))
298
299
300

```



```
data: sample(residuals(aov_region), 5000)
```

```
W = 0.87187, p-value < 2.2e-16
```

```
R squared: 0.0429
```

```
Thus Hypothesis rejected, some regions have more annual cost than others.
```

7 The Models

Applying the Simple Linear Regression we get the following graph :

```

358
359 # Correlation first: strength and direction before fitting a line.
360 cat("\nCorrelation, cost vs floor area:\n")
361 print(cor(recs$TOTSQFT_EN, recs$TOTALDOL))
362
363 lm_simple <- lm(TOTALDOL ~ TOTSQFT_EN, data = recs)
364
365 cat("\nSimple linear regression:\n")
366 print(summary(lm_simple))
367
368 # Slope in money terms.
369 slope <- coef(lm_simple)[2]

```

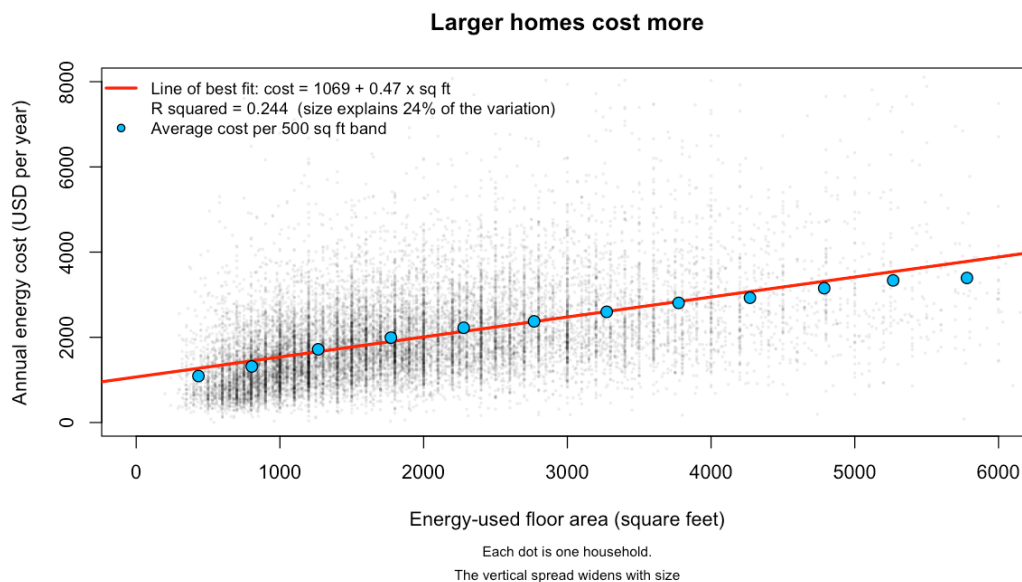
Performed well on test data but only up to 4000\$, after that the prediction is

Prediction error on the test set, USD per year:

```

> print(round(errors, 2))
              RMSE    MAE
floor area only  970.40 687.16

```

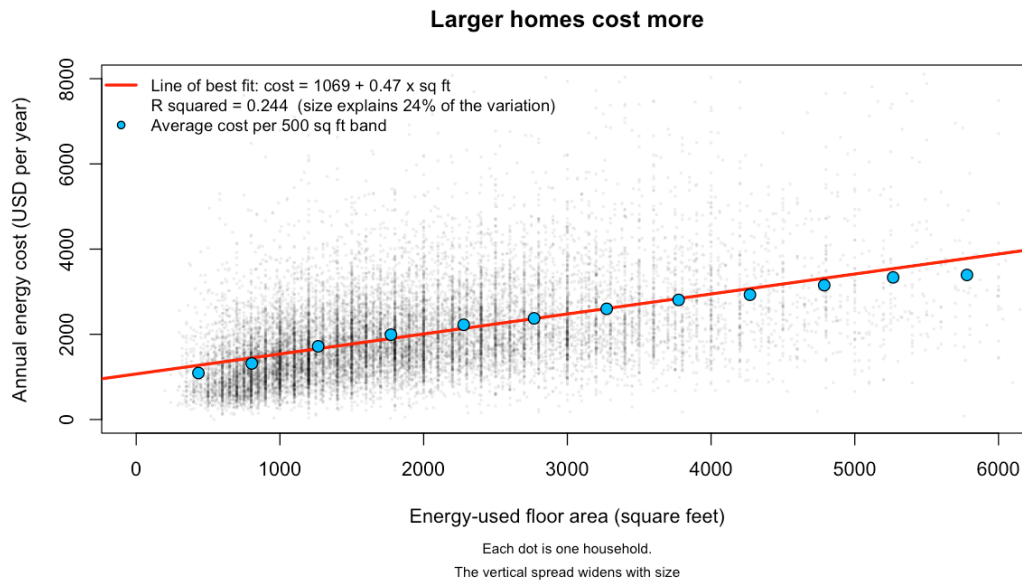


Correlation, cost vs floor area: 0.4937353

Meaning per square foot of floor area, the cost increases by ~0.50 cents per year.

Applying the Multiple Linear Regression Model

on the floor area, cooling and heating costs



```
Call:
lm(formula = TOTALDOL ~ TOTSQFT_EN + HDD65 + CDD65, data = recs)

Residuals:
    Min       1Q   Median       3Q      Max
-4835.6  -598.4  -156.9   420.0 16093.5

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.924e+02  3.765e+01  18.390  <2e-16 ***
TOTSQFT_EN    4.641e-01  6.102e-03  76.055  <2e-16 ***
HDD65         6.017e-02  5.068e-03  11.871  <2e-16 ***
CDD65         8.602e-02  1.030e-02   8.349  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 960.7 on 18491 degrees of freedom
Multiple R-squared:  0.2496,    Adjusted R-squared:  0.2495
F-statistic: 2051 on 3 and 18491 DF,  p-value: < 2.2e-16
```

Then testing it on the test split.

```
Prediction error on the test set, USD per year:

> print(round(errors, 2))
              RMSE    MAE
floor area only    970.40 687.16
plus heating/cooling 965.97 684.42
log model          1002.53 681.82

> # Outcome: climate adds little very little to the cost
```

The Root Mean Square error showed a difference of around 4.4 USD making it enough to reject the null but still adding very little to the cost overall. Thus the outcome states that Floor area is still the highest predictor of households annual energy consumption.

8 Summary

Finally all hypothesis were rejected and brought some significance to the analysis. also due to a relatively smaller dataset and its right skewed nature, the analysis is limited and hard to replicate

The data driven recommendations that can be made from this analysis for our initially mentioned stakeholders can be as following:

- Cost Reduction can be done by reducing average floor area, so creating a culture of living in not very big houses could lower overall load on utilities in specific regions where people tend to have large houses.
- Some Regions might benefit from programmes from the goverment to subsidise energy costs in order to maintain a balance of cost among regions, otherwise this disparity could lead to higher concentration of people as well as enterprises moving to lower energy cost areas.
- As for people that are looking for houses should consider buying a house that is not very large.
- Choosing an older house could be a good choice at it would generally cost less and would not affect energy cost significantly.

Furthermore, The following limitations can be observed in the analysis:

- Floor area according to the dataset is self reported, this could bring quite a bit of variance depending upon what verification measures were performed by the states when accepting these values.
- Floor area is a decent indicator but it does not tell the whole picture, a choice of only 9 out of 799 features limited the scope of this analysis, a more indepth analysis could be perform taking into consideration more of the features.

This leads us to the final part of this paper, Future work:

- Adding more useful features in the multiple regression can help the model improve and show a more clearer picture, Features like electricity only cost, cost/fuel type, etc would the model improve.
- A bigger, less skewed dataset would result in more accurate predictions where the need of log normalisation would go away and a more direct model could be built

9 Dataset and repository

Dataset: U.S. Energy Information Administration, Residential Energy Consumption Survey (RECS) 2020, Public Use Microdata, **version 7** (released January 2024). 18,496 households $\times$ 799 variables. U.S. Government work, public domain.

- Landing page: <https://www.eia.gov/consumption/residential/data/2020/index.php?view=microdata>
- CSV: https://www.eia.gov/consumption/residential/data/2020/csv/recs2020_public_v7.csv
- Codebook: <https://www.eia.gov/consumption/residential/data/2020/xls/RECS%202020%20Codebook%20for%20Public%20File%20-%20v7.xlsx>